

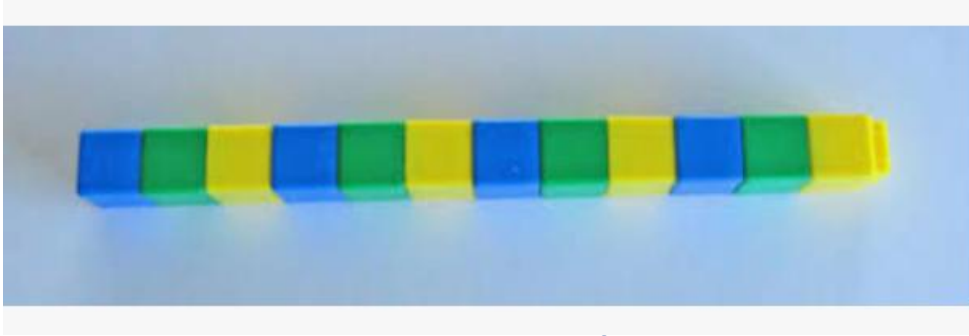
Laying Early Math Foundations: Math Talk for the Home

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Numeracy is multi-faceted



Patterning



Number Sense

(counting, more, less, addition, subtraction, etc)



Measurement



Spatial Relations

Math-related Activities

What do you do at home?

Math Workbooks



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Notifications ? Help

Shopee

math activity book

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SEARCH FILTER

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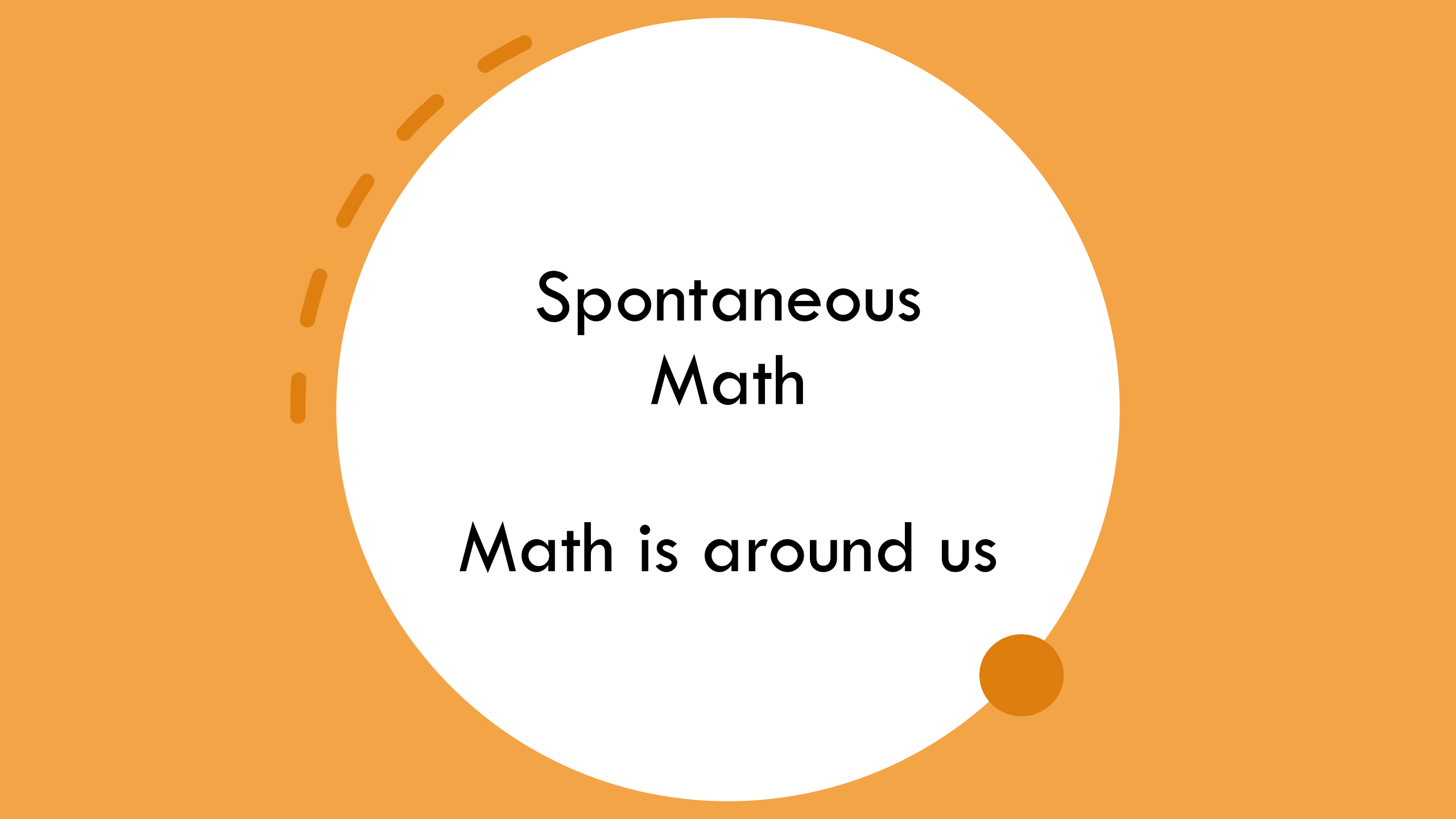
\$1 off | Any 3 enjoy 10% off

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SG

Math-related Activities

What **can** you do at home?



Spontaneous Math

Math is around us

Spotting and responding to math moments around us

Respond to what your child is asking or saying

Parent-initiated

Have you heard of these questions?

How many more stops? (on MRT, bus)

What bus are we taking?

Is this done yet? (e.g., waiting for an ad to finish, on YouTube)

Why is the ice melting?

Why do I only get 3 pieces and jeje got 4?

Parent asking: How many pieces of apple / slices of orange do you want?

How many more stops?

We got on at Clementi, now it's Commonwealth, so we just went past three stops.

We're going to Tanjong Pagar, that's five more stops.

(Use fingers!!)



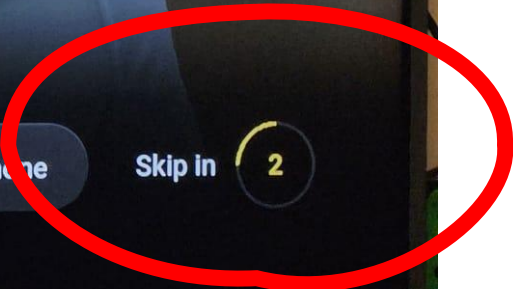


What bus are we taking?

We can take bus 48, 99 and 157
(one five seven).

Label the bus as they come
Ask them to tell you what bus it is

For triple-digit bus numbers, you
can explicitly say that the number
can also be called “one hundred
and fifty- seven”



Have you heard of these questions?

How many more stops? (on MRT, bus)

What bus are we taking?

Is this done yet? (e.g., waiting for an ad to finish, on YouTube)

Why is the ice melting?

Why do I only get 3 pieces and jeje got 4?

Parent asking: How many pieces of apple / slices of orange do you want?

Talk about Measurement (weight, distance, length)

How long
How heavy
How far
How cold
How hot



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Talk about Sizes, Shapes, Patterns

While getting dressed, talk about sizes, shapes, and patterns on clothing.

- What do you see? I see four dinosaurs, roar
- This one has a loooooong neck. This one has a short neck. What about you?
- How many red shirts do you have? Let me see. One two three.



Talk about patterns when getting dressed

Talk about Patterning

While getting dressed, talk about sizes, shapes, and patterns on clothing.

Talk about
patterns when
getting dressed



Image from Uniqlo

“Look at the pattern on your shirt!
What do you see?”

I see **narrow** blue stripe then **wide** green stripe.

- Compare to pattern on another shirt, or to sibling’s shirt

Talk about Shapes

When your child is drawing (or playing with playdoh), notice and compare different sizes.



Talk about shapes
in drawing

- Wa, so many circles. <count>
- What is the biggest/shortest/longest thing in your picture?
- Wide vs. narrow
- That tree looks so tall next to the flower!

Talk about Shapes

Describe what they / you are drawing.



Talk about shapes
in drawing

- Wa, you have many **long** lines here.
- Your circle is so **big**!
- Let mummy draw a **small/smaller one** for you
- I am going to draw **another circle** next to your circle. There are now **two circles**.

Image from <https://picklebums.com/drawings/>

Grocery shopping

Talk about
numbers in daily
routines



- We got two bags of oranges. Which one has more oranges? Which one is heavier?
- How many oranges in the bag?
- What if I eat one of the oranges?



Taking a closer look
at the “talk”

Math Talk – Spontaneous comment from child

Child gave one candy to his friend and another to his brother

C: Both of us have candies!

Other phrases and words: many, all, some, two, three, ten

Math Talk – Spontaneous comment from child

Child gave one candy to his friend and another to his brother

C: Both of us have candies!

P: So lucky.

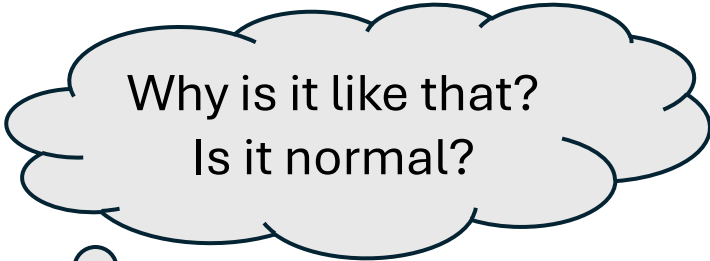
P: and you guys have two candies. One for you and one for brother

Math Talk – Following up on mistakes

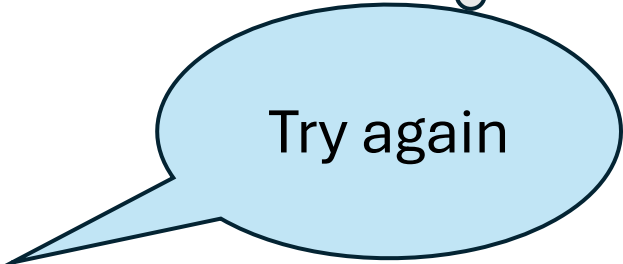
There are 4 blocks

P: How many do you have?

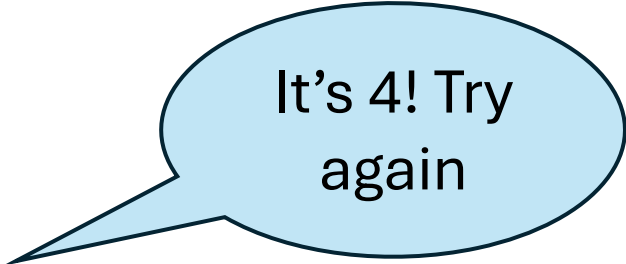
C: Six!



Why is it like that?
Is it normal?



Try again



It's 4! Try
again

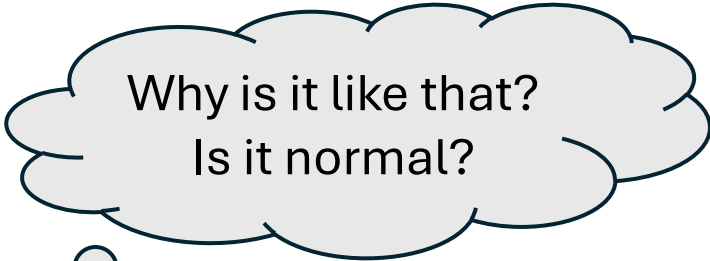
Focus on how the child
SOLVES the problem

Math Talk – Following up on mistakes

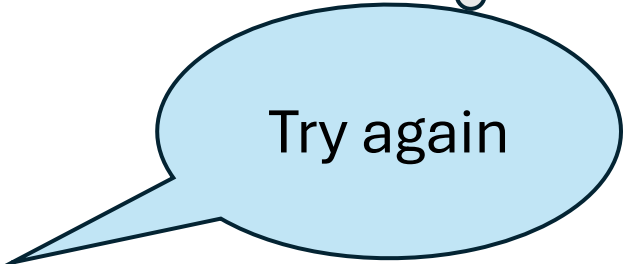
There are 4 blocks

P: How many do you have?

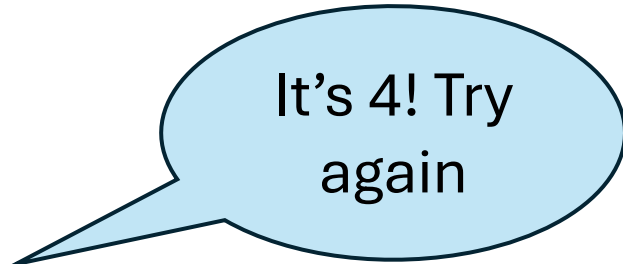
C: Six!



Why is it like that?
Is it normal?



Try again



It's 4! Try
again

There are 4 blocks

P: How many do you have?

C: Six!

P: I think I see less than 6.

P: Can you count to show
me?

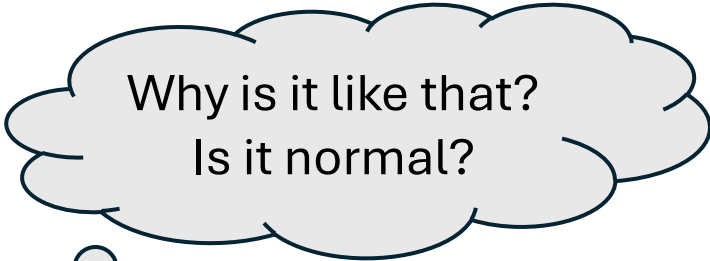
C: 1 2 3 4! [Child show four
fingers]

Math Talk – Following up on mistakes

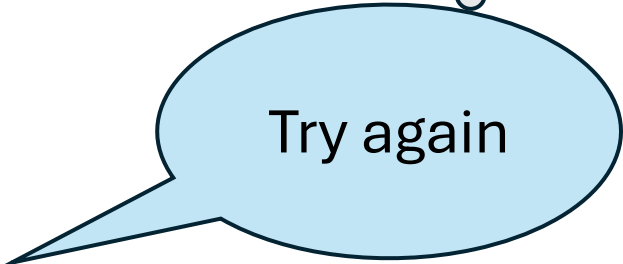
There are 4 blocks

P: How many do you have?

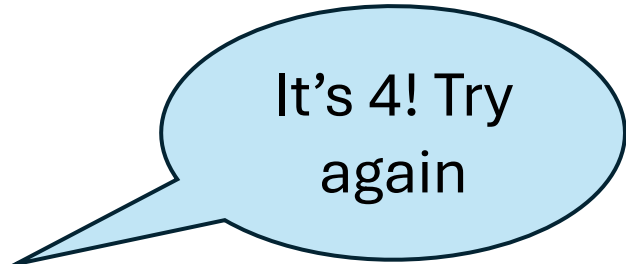
C: Six!



Why is it like that?
Is it normal?



Try again



It's 4! Try
again

There are 4 blocks

P: How many do you have?

C: Six!

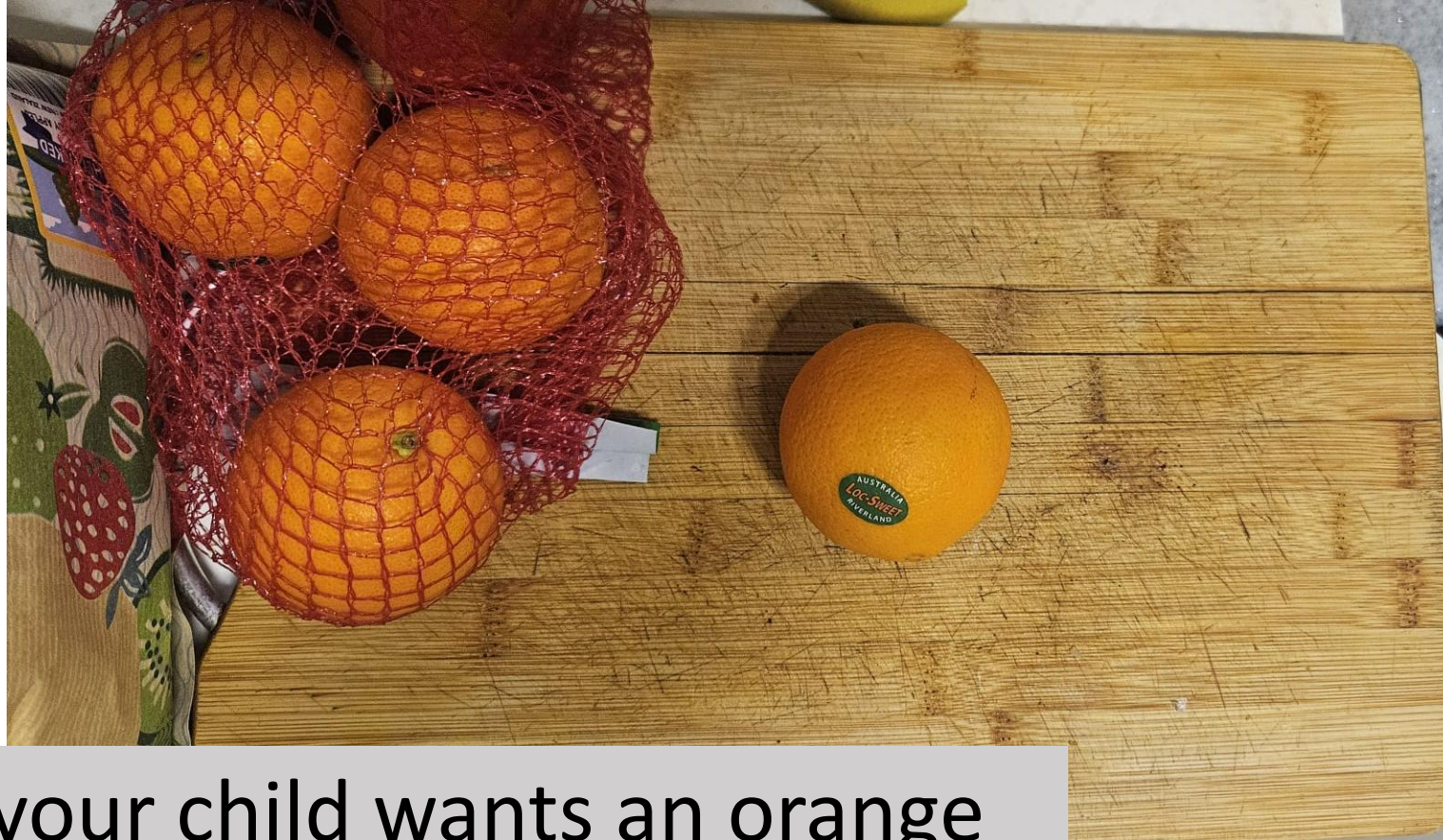
P: I think it's 4. See 1-2-3-4, 4 blocks.

Model the procedure
Model how YOU solve the
problem

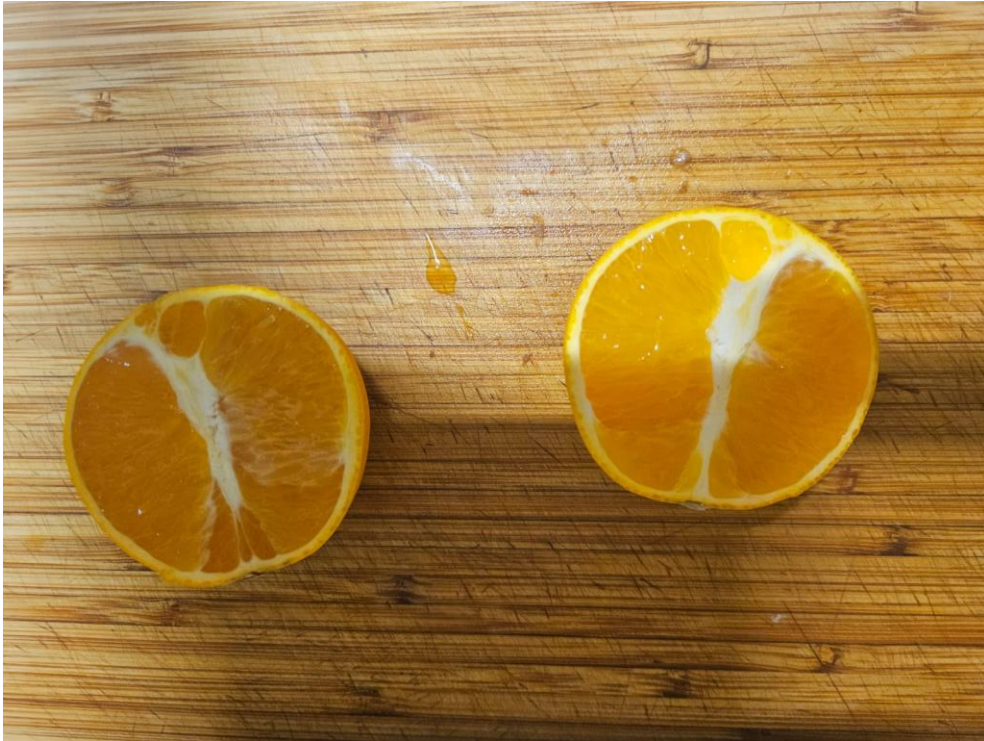




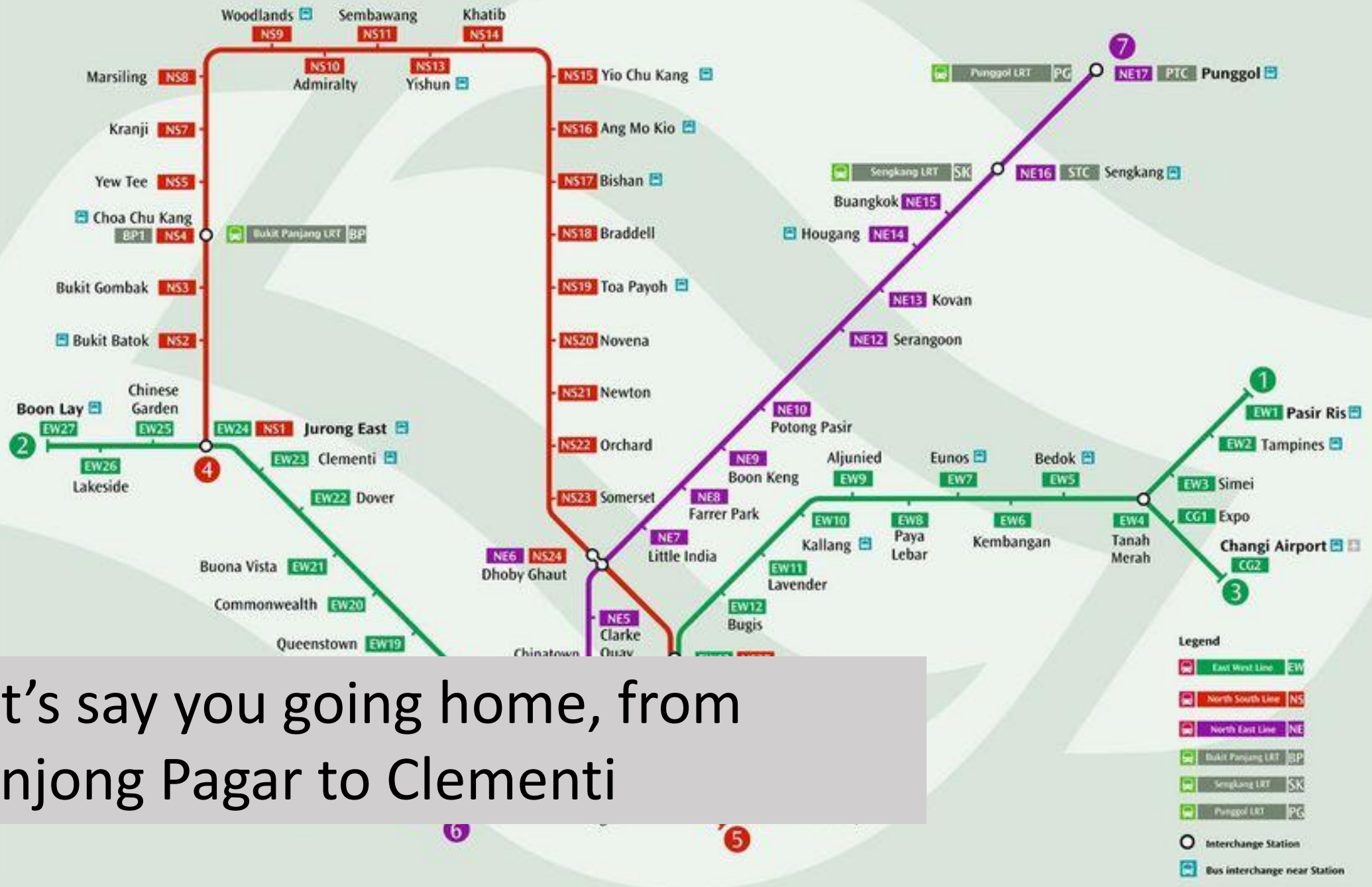
Let's try!



Let's say your child wants an orange
And you ask them to take the bag out
from the fridge



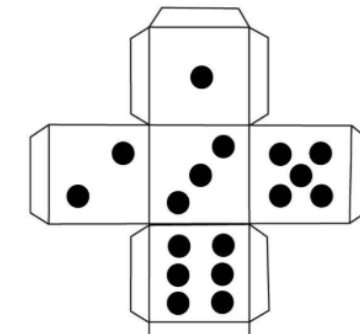
Let's say you going home, from
Tanjong Pagar to Clementi

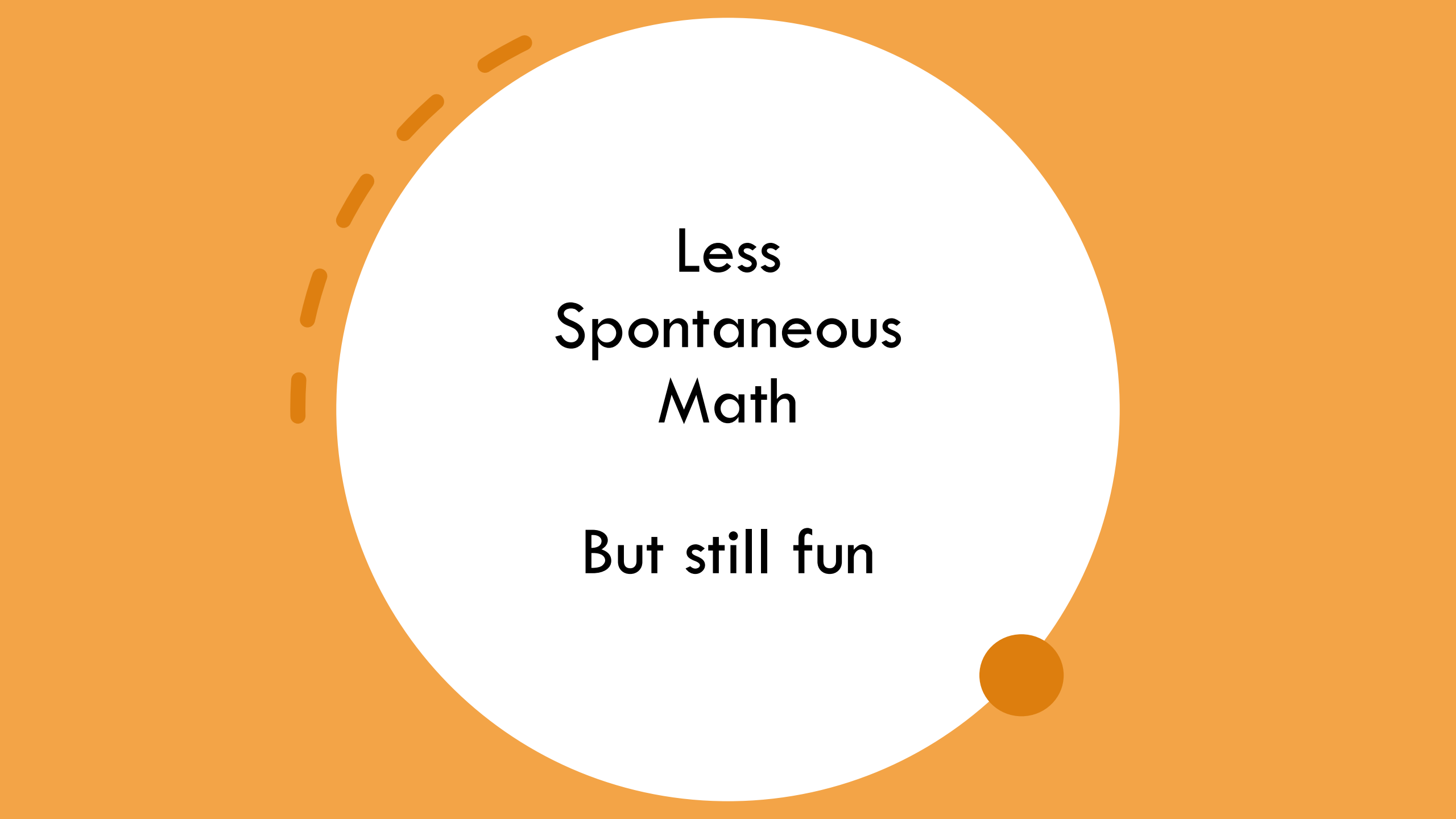




On top
Underneath
Behind
Next to
In front of

Quantities
Size
Length

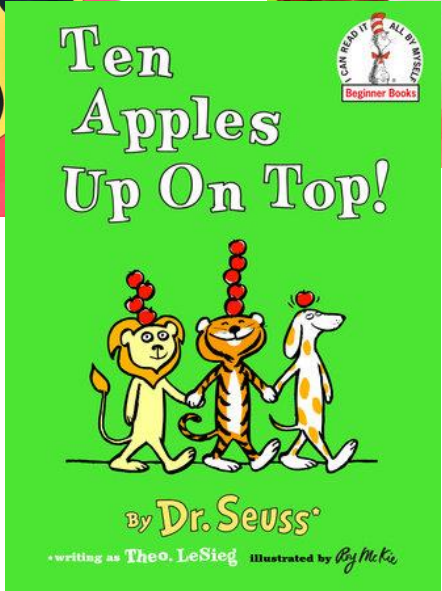
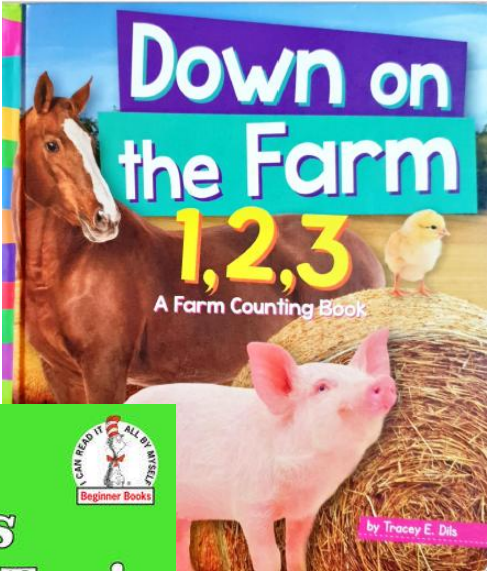
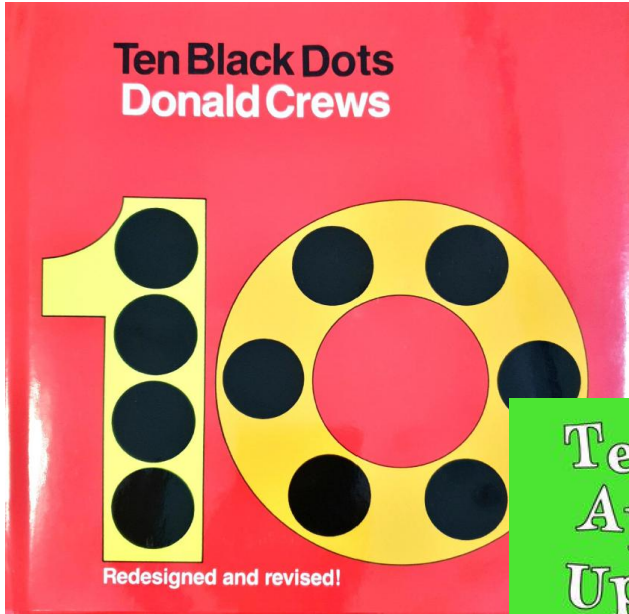




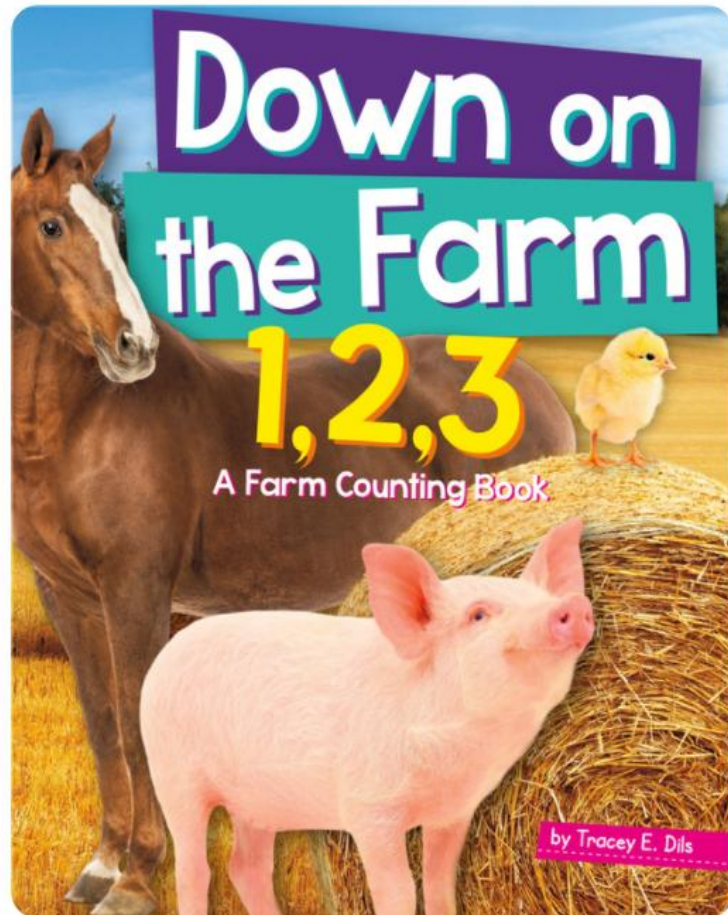
Less
Spontaneous
Math

But still fun

Intentional Math Content



Number books (non-fiction)



- We often talk about how many
- But you can ask about comparison (“more”/ “less”), about simple arithmetic (“one more”, “two more”)



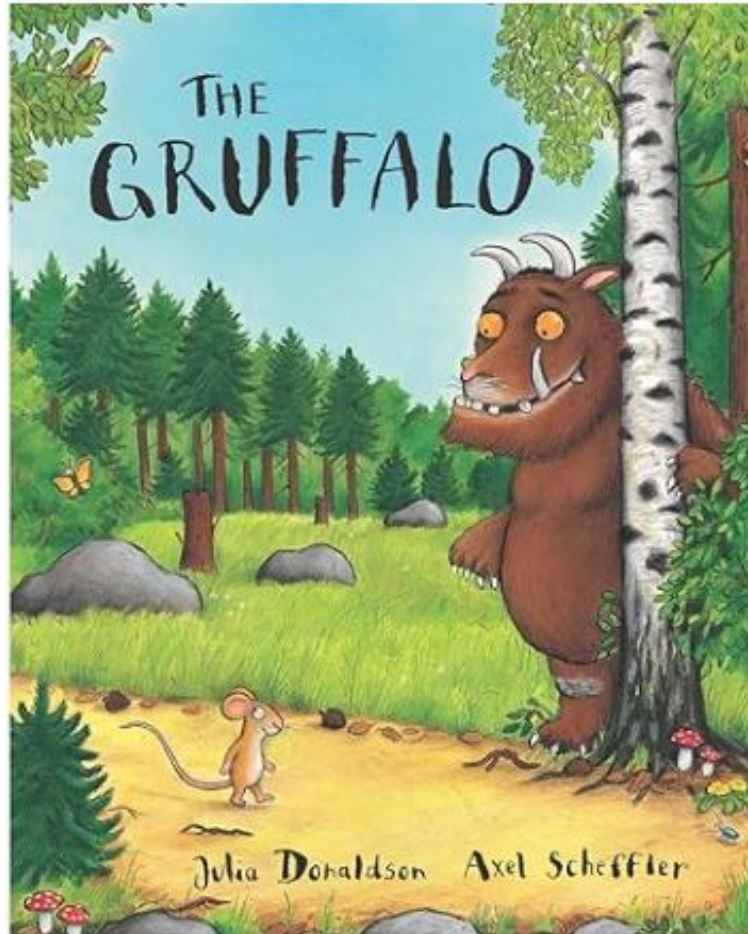
**Repeated
readings!**

Counting
Scattered
Linear

Comparison

Simple Arithmetic





- The Gruffalo saw some animals. Let's see
- First.... Second... Third.....

Let's try!

They play soccer at the park.
Someone else wants to play!





Start Small

One message

Exposure tailored to the child's environment & interest

Identify opportunities to **talk about numbers / shapes / patterns**. You just have to look around you.



Always remember!
(again, both for you as parents and for your child)

Okay to make mistake! Part of the learning process.

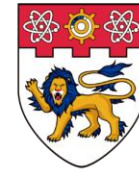
It's not about making sure you hear/get the right answer from your child or you asking the right question, but more on providing input, interactions, relationships

Thank you!

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**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE



Please reach out for questions

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